

DSA0404 – FUNDAMENTALS OF DATA SCIENCE

UNIT 3 – SOLVED EXERCISES

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1. Mean, Median, Mode and Range

Data: 12, 15, 18, 15, 20, 22, 15, 25

- Mean = 17.75
 - Median = 16.5
 - Mode = 15
 - Range = $25 - 12 = 13$
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2. Frequency Distribution

Data: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Value	Frequency
2	3
3	3
4	3
5	2
6	1

3. Q1, Q2, Q3 and IQR

Data: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Using the median-of-halves method:

- $Q1 = (8 + 10) / 2 = 9$
 - $Q2 \text{ (Median)} = (15 + 18) / 2 = 16.5$
 - $Q3 = (22 + 25) / 2 = 23.5$
 - $IQR = Q3 - Q1 = 23.5 - 9 = 14.5$
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4. Outlier Detection Using IQR

Data: 10, 12, 13, 15, 16, 17, 18, 20, 45

- $Q1 = 12.5$
- $Q3 = 19$
- $IQR = 19 - 12.5 = 6.5$

Lower boundary:

$$Q1 - 1.5(IQR) = 12.5 - 9.75 = 2.75$$

Upper boundary:

$$Q3 + 1.5(IQR) = 19 + 9.75 = 28.75$$

Since $45 > 28.75$, 45 is an outlier.

5. Skewness (Asymmetry)

Data: 5, 6, 7, 7, 8, 8, 9, 20

- Mean = 8.75
- Median = 7.5
- Mode = 7 and 8
- Mean > Median

Therefore, the distribution is positively skewed (right-skewed) due to the extreme value 20.

6. Uniform Distribution

Given:

- Minimum = 10
- Maximum = 50
- Required: $P(20 < X < 35)$

For a uniform distribution:

$$P(a < X < b) = \frac{b - a}{\text{Maximum} - \text{Minimum}}$$
$$P(20 < X < 35) = \frac{35 - 20}{50 - 10} = \frac{15}{40} = 0.375$$

Answer: 0.375 or 37.5%

7. Z-Score

Given:

- Mean = 100
- Standard deviation = 15
- $X = 130$

$$Z = \frac{X - \mu}{\sigma} = \frac{130 - 100}{15} = 2$$

8. Probability $P(X < 130)$

From Question 7:

$Z=2$

From the standard normal distribution table:

$$P(Z < 2) = 0.9772$$

$$\therefore P(X < 130) = 0.9772$$

Answer: 97.72%

9. Effect of an Extreme Value on Mean and Median

Consider:

Before outlier: 10, 12, 13, 14, 15

- Mean = 12.8
- Median = 13

Add an extreme value, 100:

After outlier: 10, 12, 13, 14, 15, 100

- Mean = 27.33
- Median = 13.5

Measure	Before	After
Mean	12.8	27.33
Median	13	13.5

Conclusion: The extreme value has a large effect on the mean, while the median remains relatively stable. Hence, the median is more robust to outliers.

10. Complete EDA on Student Marks

Consider the student marks:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 95

Central Tendency

- Mean = 62.67
- Median = 61
- Mode = No mode

Dispersion

- Minimum = 45
- Maximum = 95
- Range = 50

$$Q1 = 53.5$$

$$Q3 = 69$$

$$IQR = 69 - 53.5 = 15.5$$

Outlier Detection

Upper boundary:

$69 + 1.5(15.5) = 92.25$

Since $95 > 92.25$, 95 is an outlier.

Distribution / Skewness

Since the mean (62.67) is greater than the median (61), the dataset is positively skewed (right-skewed).

EDA Summary:

Parameter	Result
Mean	62.67
Median	61
Mode	No mode
Minimum	45
Maximum	95
Range	50
Q1	53.5
Q3	69
IQR	15.5
Outlier	95
Skewness	Positive / Right-skewed
